

Yices2 in SMT-COMP 2026

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1 Introduction

Yices2 [3] is an open-source (GPLv3) SMT solver developed and distributed by SRI International. It can be downloaded at <http://yices.csl.sri.com> and on our GitHub repository at <https://github.com/SRI-CSL/yices2>. The solver supports linear and non-linear arithmetic, bit-vectors, finite fields, uninterpreted functions, and arrays.

Yices2 uses the standard CDCL(T) [10] architecture and a variant of the Nelson-Oppen method for combining decision procedures. Details are presented in [3]. The solver also includes a Model-Construction Satisfiability Calculus (MCSat) [2, 8, 4, 7, 6] implementation. By default, MCSat is used for all theories that require non-linear arithmetic and CDCL(T) is used for everything else. Quantifier reasoning is supported for the UF theory, via E-graph matching and model-based instantiation. Yices2 can use third-party backend SAT solvers for bitvector solving. Currently, it supports three SAT solvers: CaDiCaL [1], CryptoMiniSat [11], and Kissat [1].

2 What's new?

Yices2 MCSat now includes two recent MCSat improvements: theory-specific decision heuristics [5] with target caching for better value selection, and local search integration [9] that guides the search using promising assignments. Moreover, Yices2 MCSat can now produce unsat-cores.

3 Competition Version

For SMT-COMP 2026, we are participating with the latest development version of Yices2 (<https://github.com/SRI-CSL/yices2/tree/smtcomp2026>). This version will compete in all supported logics and divisions, including the incremental, model-validation, and unsat-core tracks. Additionally, we are submitting a portfolio solver in the parallel track.

Like last year, we are utilizing Kissat as the backend SAT solver in the single-query and model-validation tracks of QF_BV.

For the QF_NIA logic, we enable the new local search feature in MCSat for both single-query and model-validation tracks.

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